

LECTURE 1 — STRESS, PATHOLOGICAL & OPEN FRACTURES (Table)

Bold = high-yield. [bracket] = standard added.

STRESS & PATHOLOGICAL FRACTURES	
Topic	Key points
Stress fracture	Repetitive stress on normal bone ; athletes, recruits, dancers, steroids, female athlete triad ; bone scan/MRI early ; rest + strengthening + brace
Pathological (general)	Biopsy = keystone ; always biopsy solitary destructive lesions
Benign tumour fractures	More common than malignant; UBC (fallen fragment sign) , NOF/fibroxanthoma (>50% diameter risk) , fibrous dysplasia (Albright's = café-au-lait + endocrinopathy)
Malignant tumour	Younger + aggressive (wide transition, periosteal reaction, night pain); biopsy, stage, chemo, resection
Metabolic	Osteoporosis, Paget's (excess bleeding, late-stage) , hyperPTH (brown tumours)
Metastatic/myeloma	2nd commonest cause; femur/humerus; renal = embolize (poor radiosensitivity) ; 61% femur, 80% peritrochanteric
Harrington criteria (prophylactic fixation)	>50% bone diameter, >2.5cm, pain after radiation, lesser trochanter fracture
Mets treatment	Adjuvant radiation (85% response) + bisphosphonates; nail/plate+cement/arthroplasty; immediate weight-bearing

OPEN FRACTURES (Gustilo)	
Aspect	Key points
Principle	All open = contaminated. 4 essentials: cover, antibiotics, debridement, stabilization
Type I	<1cm clean puncture, minimal damage, not comminuted
Type II	>1cm, no flap, moderate crush
Type IIIA/B/C	Extensive: A = bone coverable; B = periosteal stripping (flap); C = arterial injury needing repair
Antibiotics	Benzylpenicillin + flucloxacillin ×48h (+ gentamicin/metronidazole if contaminated); tetanus prophylaxis
Debridement/closure	No tourniquet; copious saline; leave open (except clean type I) → delayed primary closure at 5 days
Stabilization	External fixation safest ; IM nail (femur/tibia)

LECTURE 2 — BONE TUMOURS (Radiology Table)

*Bold = high-yield. Assess by morphology + age. Plain film best for dx; **mets** = **commonest**.*

PRIMARY MALIGNANT			
Tumour	Age	Site	X-ray
Osteosarcoma	5–20 (+ Paget's elderly)	Around knee, metaphysis	Sunburst/sunray + Codman's triangle , bone formation
Chondrosarcoma	30–50	Pelvis, scapula, femur	Lytic expansile + calcium flecks
Ewing's	5–15	Long bone diaphysis	Onion-skin periosteal reaction, permeative
GCT	20–40	Subarticular epiphysis, knee/wrist	Expanding lytic, thin cortex, locally invasive

BENIGN / TUMOUR-LIKE	
Lesion	Key points
Enchondroma	Hand, lytic + calcium flecks, pathological fracture
NOF / fibrous cortical defect	Incidental children, scalloped sclerotic margin
Fibrous dysplasia	Long bones/ribs, ground glass , well-defined
Simple bone cyst	Humerus/femur, fallen fragment sign
Aneurysmal bone cyst	Massive aneurysmal expansion , fluid levels; DDx GCT
Osteoid osteoma	Painful (night, relieved NSAIDs) , nidus <2cm + sclerotic rim , hot bone scan

MULTIPLE LESIONS & DENSITY

Topic	Key points
Metastases	Commonest; lytic (breast/bronchus/thyroid/renal/colon) ; sclerotic = men prostate, women breast
Multiple myeloma	Commonest adult primary; lytic well-defined; MRI best
Osteoporosis	↓matrix, normal mineralization; DEXA T ≤ −2.5 ; empty box/cod-fish vertebrae
Osteomalacia/rickets	Poor mineralization; Looser's zones (adult); cupped/frayed metaphysis + rachitic rosary (child)
HyperPTH	Subperiosteal resorption + brown tumours , chondrocalcinosis
↑Density	Sclerotic mets (commonest) , osteopetrosis (marble bone), myelosclerosis
Paget's	Thick trabeculae + cortex, bone enlargement, bowing , picture-frame vertebra
Haemolytic anaemia	Marrow hyperplasia → hair-on-end skull ; sickle → infarct/infection
Diaphyseal aclasia	Multiple osteochondromas, point away from joint